

**ADC0804-Controlled Infinity
Mirror with Dual-Mode
Operation
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Bench # 23
Electronics II
(EECE.3120)**

Project Description

This project is designing and building a hardware-driven infinity mirror for an electronics course project. The circuit is built around the ADC0804 Analog-to-Digital Converter, which enables analog signals to control an array of 40 LEDs arranged in an octagon. The system features dual modes: a Sensor Mode using a Light-Dependent Resistor (LDR) to react to ambient light, and a Predefined Pattern Mode using a 555 timer and op-amp integrator to generate a continuous visual “breathing” effect. The device is powered by a 9V supply regulated to 5V, ensuring stable operation under the limit of 200mA.

Final Bill of Materials

The following is the final BOM needed to complete this project.

Component	Quantity	Purpose / Notes
ADC0804 A/D Converter	1	The core of the circuit for converting the sensor's analog signal.
LM7805 Voltage Regulator	1	To create a stable 5V supply from the main 9V input for your logic chips.
555 Timer	1	To create square wave for op-amp integrator
LM358 Op-Amp	1	To build the triangle wave generator for the Predefined Pattern Mode.
Light-Dependent Resistor (LDR)	1	The sensor that will detect the ambient light level for the Sensor Mode.
SPDT Slide Switch	1	To switch the ADC's input between the LDR circuit and the triangle wave generator.
LEDs (5mm Green)	40	The visual part of the mirror.
NPN Transistors (2N3904)	8	To function as switches, allowing each of the 8 ADC outputs to safely drive a group of 6 LEDs.

Component	Quantity	Purpose / Notes
Resistors	Various	1k (41) 1.5k (9) 10k (3) 47k (2)
Capacitors	Various	For power supply filtering on the ADC0804 and LM7805, and for the oscillator circuit. Triangle Wave Generator 22u non-polar (1) 1u non-polar (1) 0.1u non-polar (2) 0.01u non-polar (1) 0.1u polar (1) 47u polar (1)
DIP Sockets	3	Safely house the ICs for easy replacement 20-pin DIP socket (1) 8-pin DIP socket (2)
4-inch Round Mirror	1	Supplied by the course.
4-inch Round Plexiglass	1	Supplied by the course.
Reflective Film	1	Supplied by the course.
Project Enclosure	1	A professional-style project box is required.

Component	Quantity	Purpose / Notes
22-gauge Wire	~24 inches each	Red for the +9V input and black for ground.

Circuit Description

The circuit is designed to control an 8-sided infinity mirror display using an ADC0804 analog-to-digital converter. The system operates in two distinct modes, selectable by a slide switch: a sensor-driven mode and a predefined pattern mode. The entire system is powered by a 9V DC source, which is regulated to a stable 5V DC using an LM7805 voltage regulator. This 5V rail powers the logic components, including the ADC0804, the 555 timer, and the LED arrays.

Input Stage: Dual-Mode Selection

The core of the input stage is an SPDT slide switch that toggles the input to the ADC between two sub-circuits:

1. *Sensor Mode:* This mode uses a Light-Dependent Resistor (LDR) configured in a voltage divider circuit. As the ambient light intensity in the room changes, the resistance of the LDR shifts, creating a varying analog voltage. This allows the mirror to react dynamically to its environment.
2. *Predefined Pattern Mode:* This mode uses an internal waveform generator to create a continuous visual effect. A 555 timer is configured as an astable multivibrator to generate a square wave with a frequency of approximately 215 Hz. This square wave is fed into an LM358 op-amp configured as an integrator, which converts the signal into a continuous triangle wave centered around a 2.5V reference voltage.

Signal Processing: Analog-to-Digital Conversion

The selected analog signal is fed into the pin of the ADC0804 8-bit A/D converter. The ADC samples this voltage and converts it into a corresponding 8-bit digital output (D0–D7). The ADC is set to a free-running mode, continuously updating the digital output as the input voltage changes.

Output Stage: LED Driver

The visual display consists of 40 green LEDs arranged in 8 groups of five, matching the octagonal shape of the mirror frame. Since ADC0804 cannot supply sufficient current to drive the LEDs directly, the digital outputs are used to control 8 NPN transistors (2N3904).

Each transistor acts as a low-side switch for one group of 5 LEDs. When a digital output pin goes HIGH, it activates the corresponding transistor, completing the circuit to ground and illuminating

that specific side of the octagon. To ensure the total current draw remains below the project limit of 200 mA, each individual LED is wired with a 1kohm current-limiting resistor, resulting in a safe operating current of approximately 2.9 mA per LED.

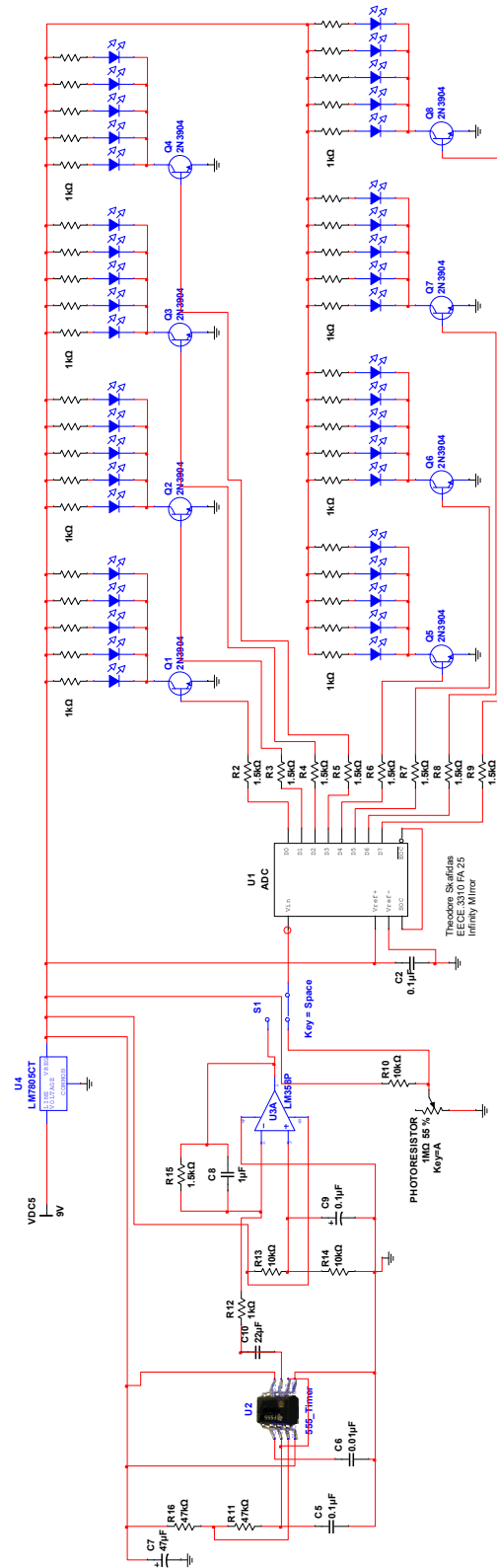


Figure 1 – Full Led ADC Circuit

Circuit Analysis

The design ensures component safety and functionality through the following calculations:

LED Current Limiting

To adhere to the project's strict 200 mA current limit, the current through the LEDs was minimized. The LEDs are powered by the 5V rail with a forward voltage (V_f) of approximately 2.1V. A 10k Ω resistor was selected for each LED.

$$I_{LED} = \frac{V_{CC} - V_f}{R} = \frac{5V - 2.1V}{10k} = 2.9mA$$

With 40 LEDs active simultaneously:

$$I_{Total\ LEDs} = 40 * 2.9mA = 116mA$$

Adding the quiescent current of the ICs (~10-20 mA), the total draw remains safely below 200 mA.

Predefined Pattern Frequency (555 Timer)

The 555 timer operates in astable mode to generate the clock signal for the integrator. The frequency is determined by R_1 (47k Ω), R_2 (10k Ω), and C (0.1 μ F).

$$f = \frac{1.44}{(R_1 + 2R_2)C} = \frac{1.44}{(47k + 2(10k)) * 0.1\mu F} = 215Hz$$

This frequency of 215 Hz was chosen to work with the op-amp integrator to create a smooth triangle wave, allowing the ADC to cycle through its output range rapidly enough to create a "solid bar" visual effect due to the persistence of vision.

Triangle Wave Generator (Integrator)

The LM358 op-amp is configured as an integrator. The 22 μ F coupling capacitor removes the DC offset from the 555 timer's output, centering the signal around the 2.5V reference established by the voltage divider (R_{13} and R_{14}). This ensures the triangle wave stays within the 0V–5V input range of the ADC.

Measurement Results

Simulation Results

The circuit was first verified in Multisim. The triangle wave generator successfully produced a wave oscillating between approximately 1.6V and 3.5V.

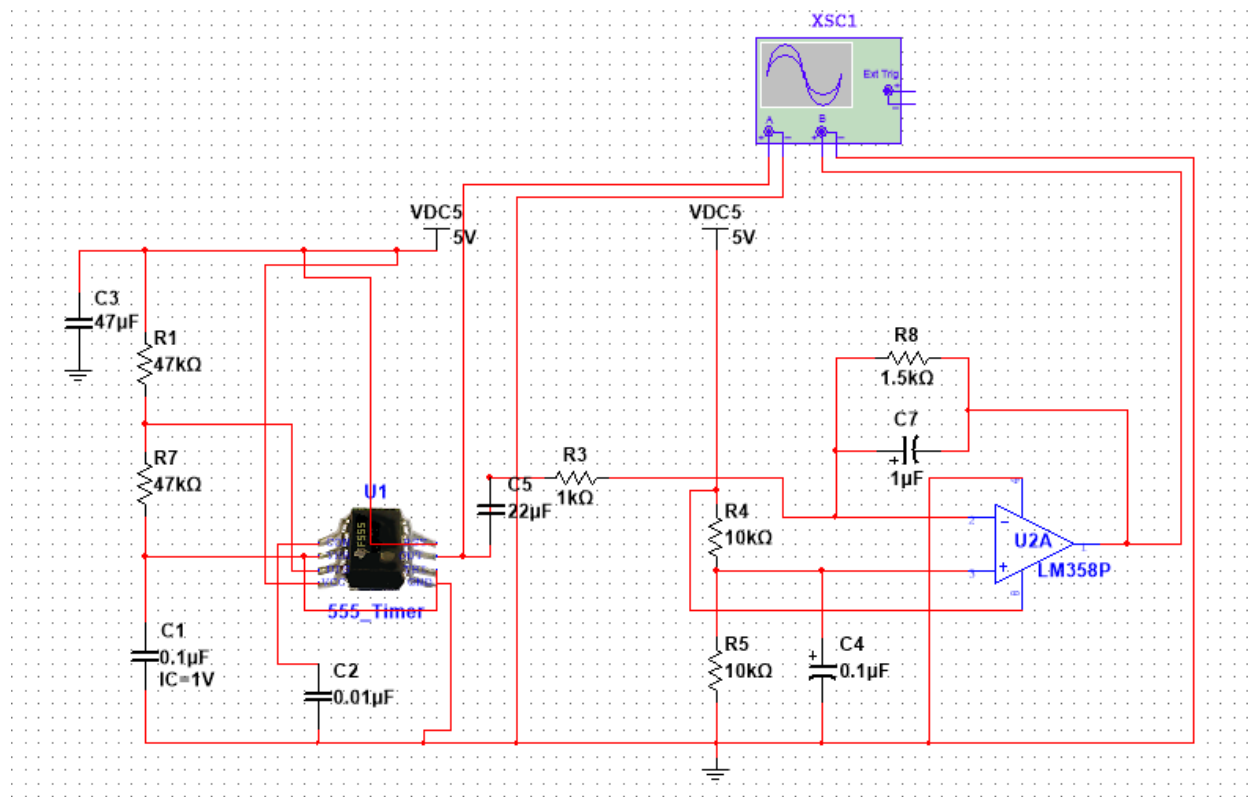


Figure 2 – Triangle Wave Generator

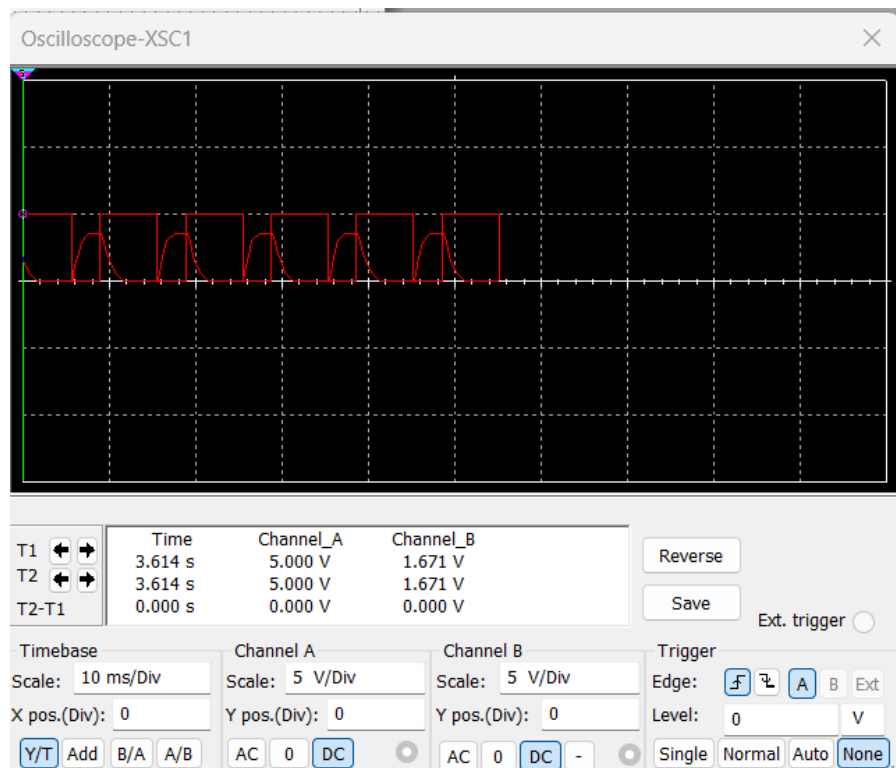


Figure 3 – Triangle Wave Generator Output

Physical Test Results

Prior to final assembly, the LDR and Triangle wave circuits were tested on a breadboard. The LM7805 provided a steady 4.98V output from the 9V input. The total current draw of the finished device was measured to be around 124mA, which verifies the theoretical calculations. However, **the final soldered assembly is currently non-functional**. The LED array does not light up in either Sensor Mode or Predefined Pattern Mode.

Discussion

During the construction phase, several design choices and challenges were addressed:

Component Selection

Originally, 220ohm resistors were considered for the LEDs. However, calculation showed this would exceed the 200mA limit. Switching to 1kohm resistors reduced the current to 2.9mA per LED.

Troubleshooting

During the initial testing of the LM7805 regulator, a short circuit occurred due to a wiring error on the TO-220 package. This was resolved by verifying the pinout and isolating the component.

Fabrication and Assembly

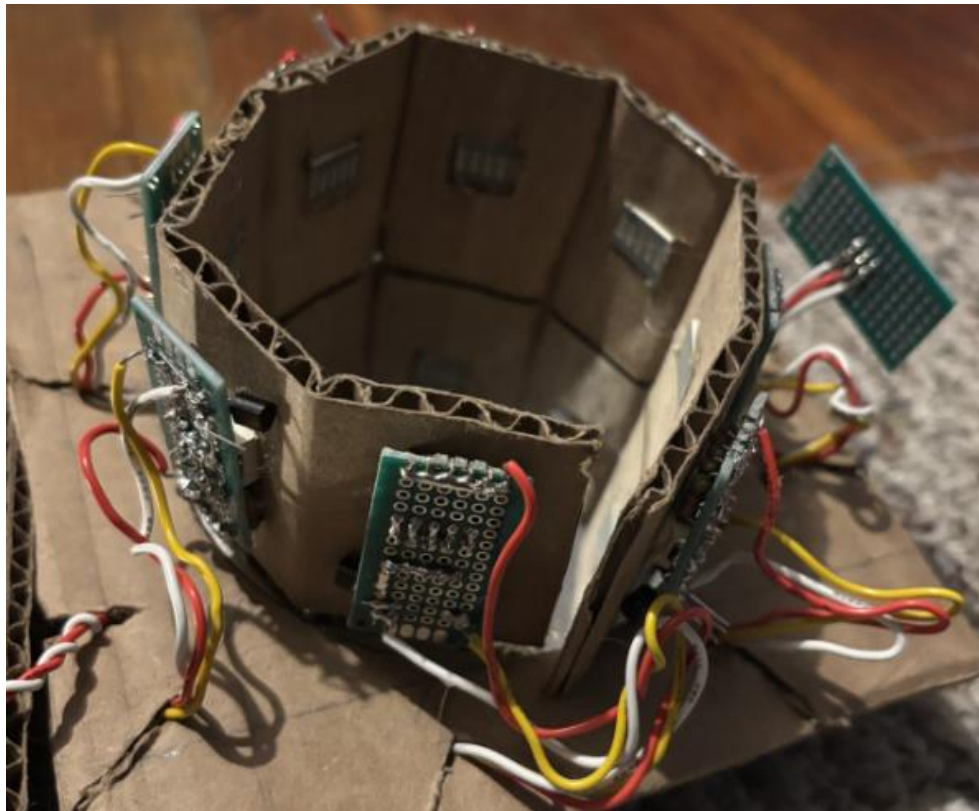
A critical design improvement was identified and implemented prior to soldering: the use of DIP sockets for all Integrated Circuits (ICs). This ensured that the sensitive chips (ADC0804, 555, LM358) were not exposed to soldering heat, protecting them from thermal damage and allowing for easy removal.

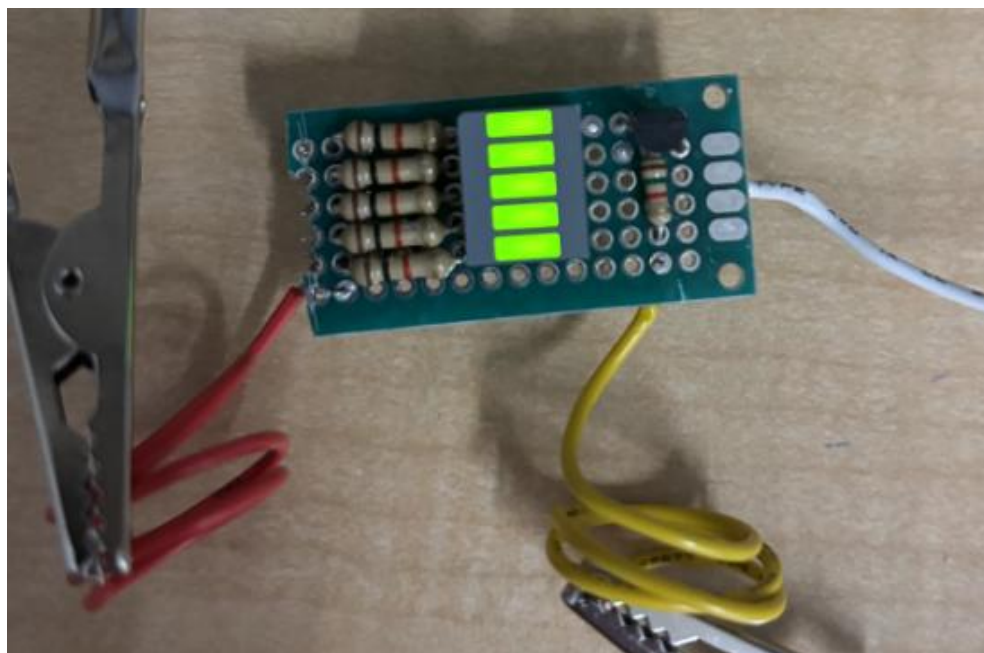
Assembly Failure

Despite the protection provided by the sockets, **the final device is non-functional**. Verification of the circuit connections was not completed during the soldering process. Consequently, it is suspected that an incomplete connection (open circuit) exists somewhere on the main board traces, preventing the signal from reaching the LED drivers. Because the chips are known to be safe (thanks to the sockets), the failure is isolated entirely to the board's physical wiring.

Conclusion

The project successfully demonstrated the theoretical design and simulation of an ADC0804-controlled infinity mirror. The "Dual-Mode" concept was verified in simulation and initial breadboard tests. The final physical build, however, did not meet the functional requirements due to a manufacturing error on the protoboard. While the decision to use DIP sockets successfully prevented thermal damage to the logic chips, the device failed to illuminate due to a suspected open circuit in the soldered connections. This underscores the necessity of continuous continuity testing during the soldering phase, even when component safety is assured.





References

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